

Study Notes: Determinants

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Determinant of a 2×2 matrix

Consider

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

The determinant of A is

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

Determinant of a 3×3 matrix

Let

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$$

A cofactor expansion along the first row gives

$$\det(A) = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}.$$

Using the 2×2 formula inside, this becomes

$$\det(A) = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}).$$

Equivalently (Sarrus' rule):

$$\det(A) = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}.$$

Determinant of a 4×4 matrix

Let

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}.$$

The cofactor C_{1j} of the entry a_{1j} is

$$C_{1j} = (-1)^{1+j} M_{1j},$$

where M_{1j} is the determinant of the 3×3 minor obtained by deleting row 1 and column j .

A cofactor expansion along the first row is

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} + a_{14}C_{14}.$$

Explicitly,

$$\det(A) = a_{11} \begin{vmatrix} a_{22} & a_{23} & a_{24} \\ a_{32} & a_{33} & a_{34} \\ a_{42} & a_{43} & a_{44} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} & a_{24} \\ a_{31} & a_{33} & a_{34} \\ a_{41} & a_{43} & a_{44} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} & a_{24} \\ a_{31} & a_{32} & a_{34} \\ a_{41} & a_{42} & a_{44} \end{vmatrix} - a_{14} \begin{vmatrix} a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \end{vmatrix}.$$

Each 3×3 determinant can be expanded using the 3×3 formulas above.

Useful properties

For any $n \times n$ matrix A :

- $\det(A) = 0$ if and only if A is singular (not invertible).
- Swapping two rows multiplies the determinant by -1 .
- Multiplying a row by a scalar k multiplies the determinant by k .
- Adding a multiple of one row to another row does not change the determinant.
- $\det(AB) = \det(A)\det(B)$ for any $n \times n$ matrices A, B .
- $\det(A^T) = \det(A)$.

Examples↓

Example 1: 2×2 determinant

Compute the determinant of

$$A = \begin{bmatrix} 2 & -3 \\ 4 & 1 \end{bmatrix}.$$

Using the formula $\det(A) = ad - bc$, we have

$$\det(A) = (2)(1) - (-3)(4)$$

$$\det(A) = 2 - (-12)$$

$$\det(A) = 2 + 12 = 14.$$

Example 2: 3×3 determinant (cofactor expansion)

Compute the determinant of

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 4 \\ 2 & 1 & 0 \end{bmatrix}.$$

Expand along the first row:

$$\det(B) = 1 \begin{vmatrix} -1 & 4 \\ 1 & 0 \end{vmatrix} - 2 \begin{vmatrix} 0 & 4 \\ 2 & 0 \end{vmatrix} + 3 \begin{vmatrix} 0 & -1 \\ 2 & 1 \end{vmatrix}.$$

Now evaluate each 2×2 determinant:

$$\begin{vmatrix} -1 & 4 \\ 1 & 0 \end{vmatrix} = (-1)(0) - (4)(1) = -4,$$

$$\begin{vmatrix} 0 & 4 \\ 2 & 0 \end{vmatrix} = (0)(0) - (4)(2) = -8,$$

$$\begin{vmatrix} 0 & -1 \\ 2 & 1 \end{vmatrix} = (0)(1) - (-1)(2) = 2.$$

Substitute back:

$$\det(B) = 1(-4) - 2(-8) + 3(2)$$

$$\det(B) = -4 + 16 + 6$$

$$\det(B) = 18.$$

Example 3: 3×3 determinant (Sarrus' rule)

Compute the determinant of

$$C = \begin{bmatrix} 2 & 0 & 1 \\ -1 & 3 & 2 \\ 4 & -2 & 1 \end{bmatrix}.$$

Using Sarrus' rule, write the first two columns again:

$$\begin{array}{ccc|cc} 2 & 0 & 1 & 2 & 0 \\ -1 & 3 & 2 & -1 & 3 \\ 4 & -2 & 1 & 4 & -2 \end{array}$$

Sum of downward diagonals:

$$(2)(3)(1) + (0)(2)(4) + (1)(-1)(-2) = 6 + 0 + 2 = 8.$$

Sum of upward diagonals:

$$(1)(3)(4) + (2)(2)(2) + (0)(-1)(1) = 12 + 8 + 0 = 20.$$

Therefore,

$$\det(C) = 8 - 20 = -12.$$

Example 4: 4×4 determinant with zeros

Compute the determinant of

$$D = \begin{bmatrix} 1 & 0 & 2 & -1 \\ 3 & 1 & 0 & 2 \\ 0 & -2 & 1 & 0 \\ 2 & 0 & 1 & 1 \end{bmatrix}.$$

Expand along the first row (because it has a zero):

$$\det(D) = 1 \cdot C_{11} + 0 \cdot C_{12} + 2 \cdot C_{13} + (-1) \cdot C_{14}.$$

So

$$\det(D) = 1 \cdot \begin{vmatrix} 1 & 0 & 2 \\ -2 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} + 2 \cdot \begin{vmatrix} 3 & 1 & 2 \\ 0 & -2 & 0 \\ 2 & 0 & 1 \end{vmatrix} - 1 \cdot \begin{vmatrix} 3 & 1 & 0 \\ 0 & -2 & 1 \\ 2 & 0 & 1 \end{vmatrix}.$$

Call these three 3×3 determinants D_1 , D_2 , and D_3 .

Compute D_1 :

$$D_1 = \begin{vmatrix} 1 & 0 & 2 \\ -2 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix}.$$

Expand along the first row:

$$D_1 = 1 \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} - 0 \begin{vmatrix} -2 & 0 \\ 0 & 1 \end{vmatrix} + 2 \begin{vmatrix} -2 & 1 \\ 0 & 1 \end{vmatrix}.$$

Compute the 2×2 determinants:

$$\begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = (1)(1) - (0)(1) = 1,$$

$$\begin{vmatrix} -2 & 1 \\ 0 & 1 \end{vmatrix} = (-2)(1) - (1)(0) = -2.$$

Thus

$$D_1 = 1(1) + 2(-2) = 1 - 4 = -3.$$

Compute D_2 :

$$D_2 = \begin{vmatrix} 3 & 1 & 2 \\ 0 & -2 & 0 \\ 2 & 0 & 1 \end{vmatrix}.$$

Expand along the second row (it has a zero):

$$D_2 = 0 \cdot \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} - (-2) \cdot \begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix} + 0 \cdot \begin{vmatrix} 3 & 1 \\ 2 & 0 \end{vmatrix}.$$

Only the middle term remains:

$$D_2 = 2 \begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix}.$$

Compute the 2×2 determinant:

$$\begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix} = (3)(1) - (2)(2) = 3 - 4 = -1.$$

So

$$D_2 = 2(-1) = -2.$$

Compute D_3 :

$$D_3 = \begin{vmatrix} 3 & 1 & 0 \\ 0 & -2 & 1 \\ 2 & 0 & 1 \end{vmatrix}.$$

Expand along the first row (because of the zero):

$$D_3 = 3 \begin{vmatrix} -2 & 1 \\ 0 & 1 \end{vmatrix} - 1 \begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} + 0 \cdot \begin{vmatrix} 0 & -2 \\ 2 & 0 \end{vmatrix}.$$

Compute the 2×2 determinants:

$$\begin{vmatrix} -2 & 1 \\ 0 & 1 \end{vmatrix} = (-2)(1) - (1)(0) = -2,$$

$$\begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} = (0)(1) - (1)(2) = -2.$$

Thus

$$D_3 = 3(-2) - 1(-2) = -6 + 2 = -4.$$

Combine to get $\det(D)$:

$$\det(D) = 1 \cdot D_1 + 2 \cdot D_2 - 1 \cdot D_3$$

$$\det(D) = (-3) + 2(-2) - (-4)$$

$$\det(D) = -3 - 4 + 4 = -3.$$

So

$$\det(D) = -3.$$